Use of STEP AP239 PLCS to enable collaboration across disciplines and enterprises

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My name is Nigel Shaw

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- Chair of Editing Committee for STEP initial release
- Chair of ProSTEP Round Table
- Eurostep project lead on CRESCENDO and TOICA

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» How does it fit into the PDT Europe 2014 theme?

Product Life Cycle Support (PLCS)

ISO 10303-239

- A joint industry and government initiative to accelerate development of new standards for product support information
- An international project to produce an approved ISO standard within 4 years
  - Commenced November 1999
  - PLCS, Inc closed down 2004
  - Standard published in 2005
  - Edition 2 published in 2012
- PLCS will ensure support information is aligned to the evolving product definition over the entire life cycle
- PLCS extends ISO 10303 STEP - the STandard for Exchange of Product model data

PLCS Background and Design Objectives

How to keep the information needed to operate and maintain a product aligned with the changing product over its life cycle in a heterogeneous organization, process and system environment?

Product Definition Information

Maintenance Schedules

Tools

Test Equipment

Transportation

Consumables

Software

Spares

Training

Support Facilities

Storage Requirements

Product Life Cycle Support (PLCS)

Membership – Nov 2002

Finnish Defence Forces

FMV

DEPARTMENT OF DEFENSE UNITED STATES OF AMERICA

SAAB TECHNOLOGIES

FORSVARSDEPARTEMENTET
The Royal Norwegian Ministry of Defence

BOEING

BAE SYSTEMS

HÄGGLUNDS VEHICLE

DNV
DET NORSKE VERITAS

LOCKHEED MARTIN

Rolls-Royce

AEROSYSTEMS INTERNATIONAL

BaaN

PTC

LSC GROUP

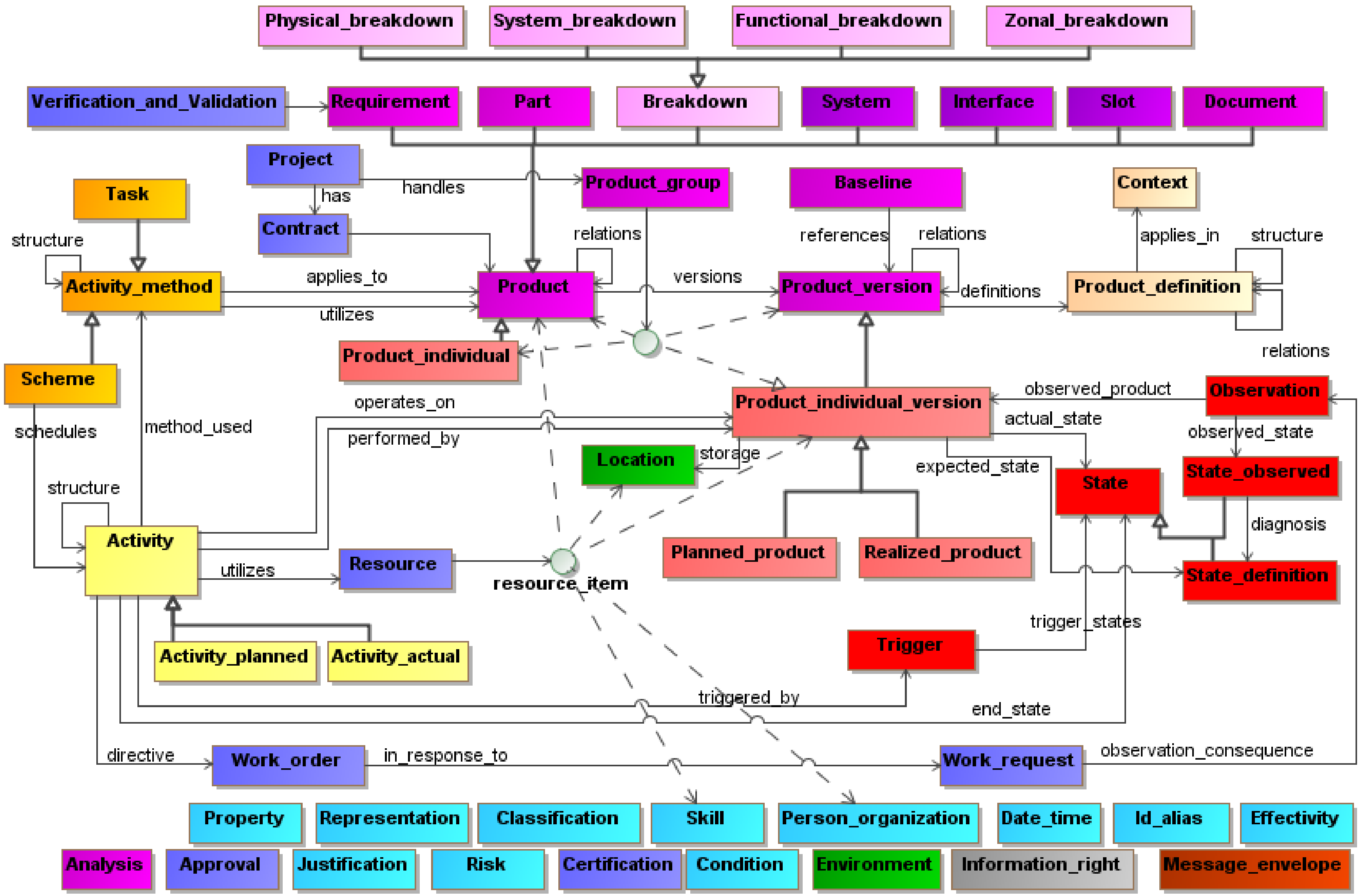
Pennant

EDS

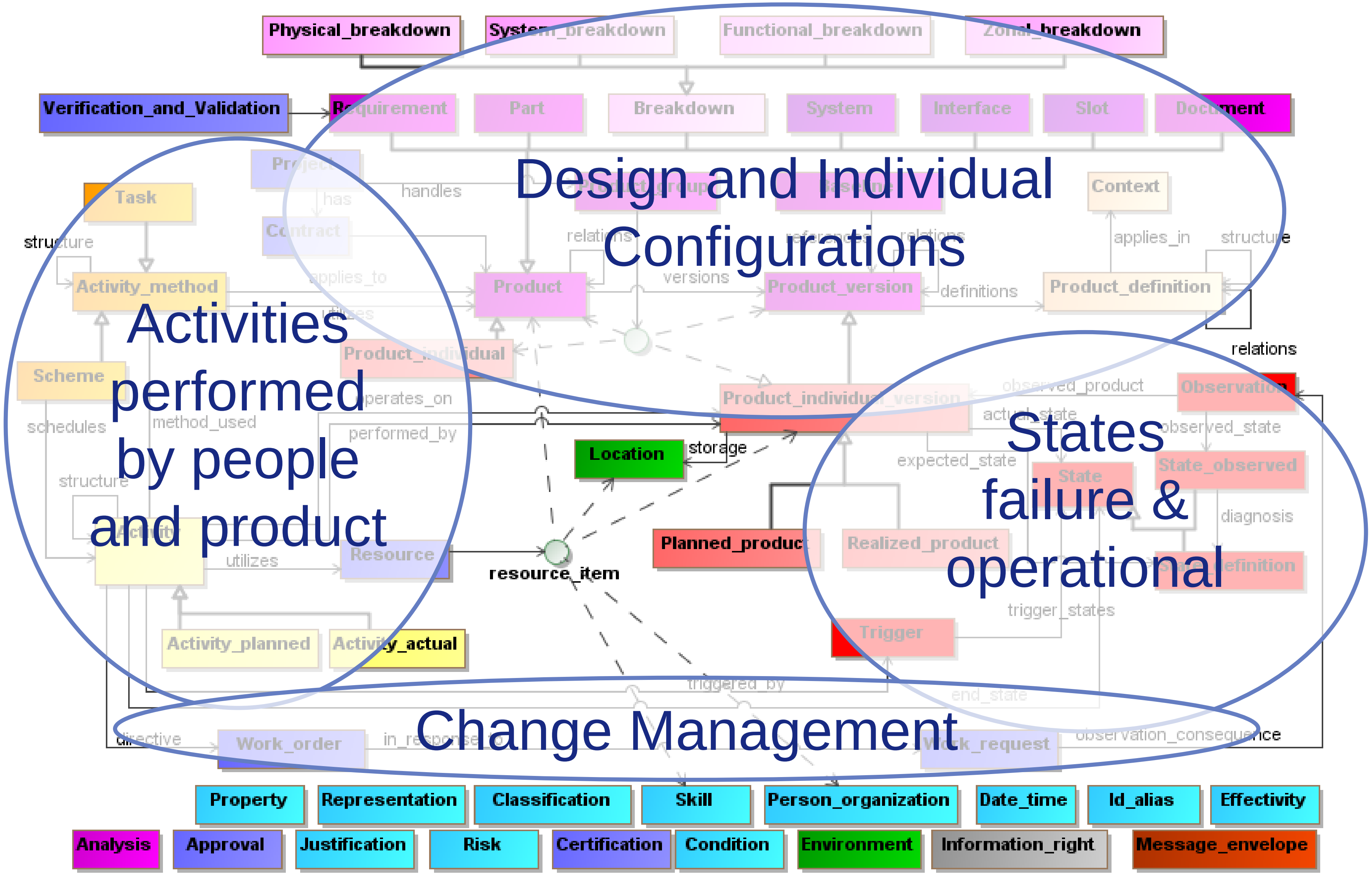
INDUSTRIAL & FINANCIAL SYSTEMS IFS

Supported by: ATI and eurostep

The PLCS Concept model - A high level view of PLCS content



The PLCS Histories

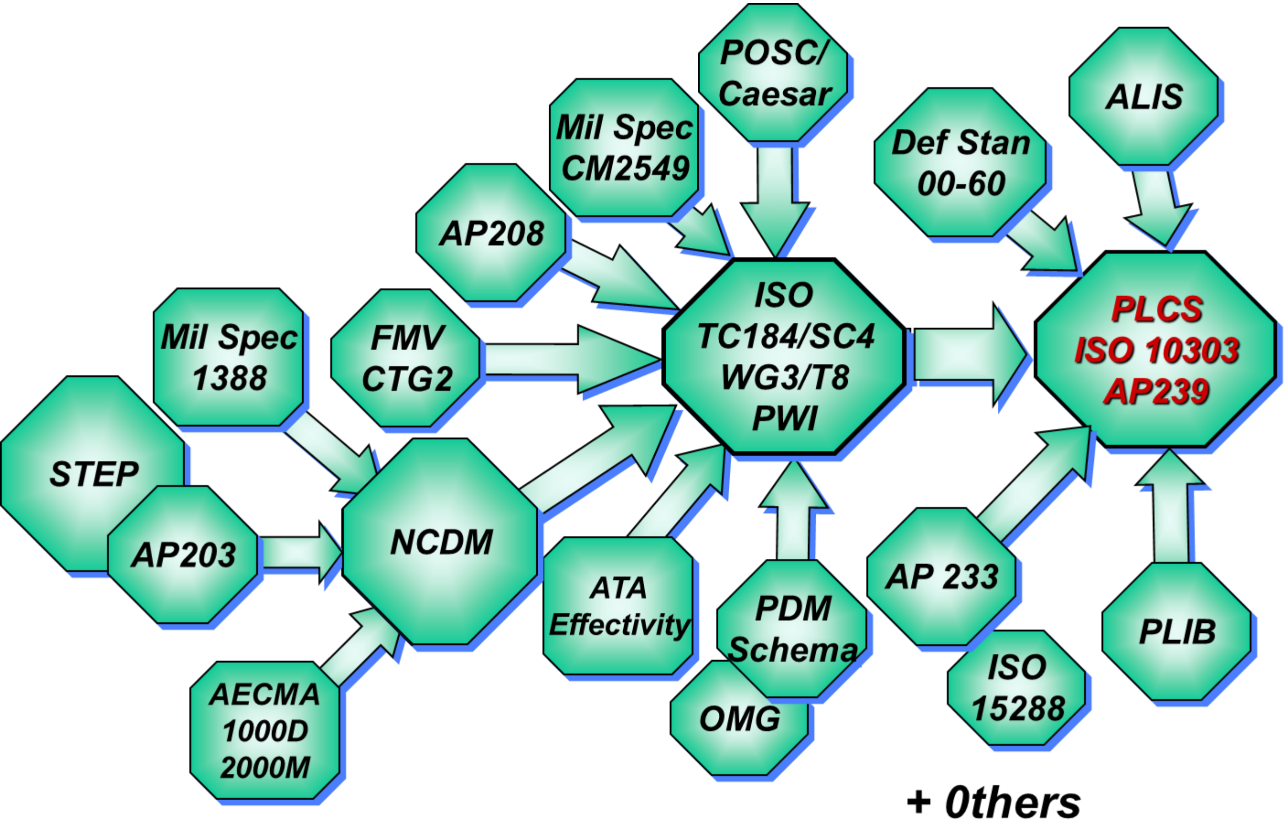


Design Objectives

- To avoid encoding business rules
- To enable incremental growth of data over the life cycle
  - Has to support a data warehouse perspective, not just the exchange of snapshots
- To allow for but not require complexity
  - PLCS is a large model that reflects the needed capabilities
  - Individual areas can be handled in a minimal fashion
    - Data Exchange Specifications (DEXs)
- To support the legacy approaches (standards)
  - Enable smarter more joined up approaches too!

Relationship to other standards

- Inputs



- Overlaps
  - AP233 for Systems Engineering
  - AP242 through harmonisation
- Areas not explicitly covered but relevant
  - OAGIS and related transaction standards
  - MIMOSA
    - Health and Usage Monitoring
  - Geometry
    - Supports relative positioning
    - Use AP203/214/242 or Native or JT
  - OSLC
    - How versus What – can be used together

How it can applied and why it works?

- Eurostep started on Share-A-space™ in late 1999 based on STEP AP214
- Eurostep's implementations of PLCS is connected to Share-A-space™
- PLCS is one of the main ways to input into Share-A-space™
  - Consolidation is based on capabilities from PLCS

Kongsberg Protech Systems

- PROTECTOR - RWS – Medium Caliber
- 8 offices in 4 countries
- 800 employees (approx.)
- Revenues MNOK 4 185 (2011)
- Soldier protection and survivability
- Enhanced situational awareness
- 17 countries have chosen RWS

The world’s leading supplier of Remote Weapon Stations

KPS Business Collaboration – Product Life Cycle Support

KONGSBERG

The main goals:

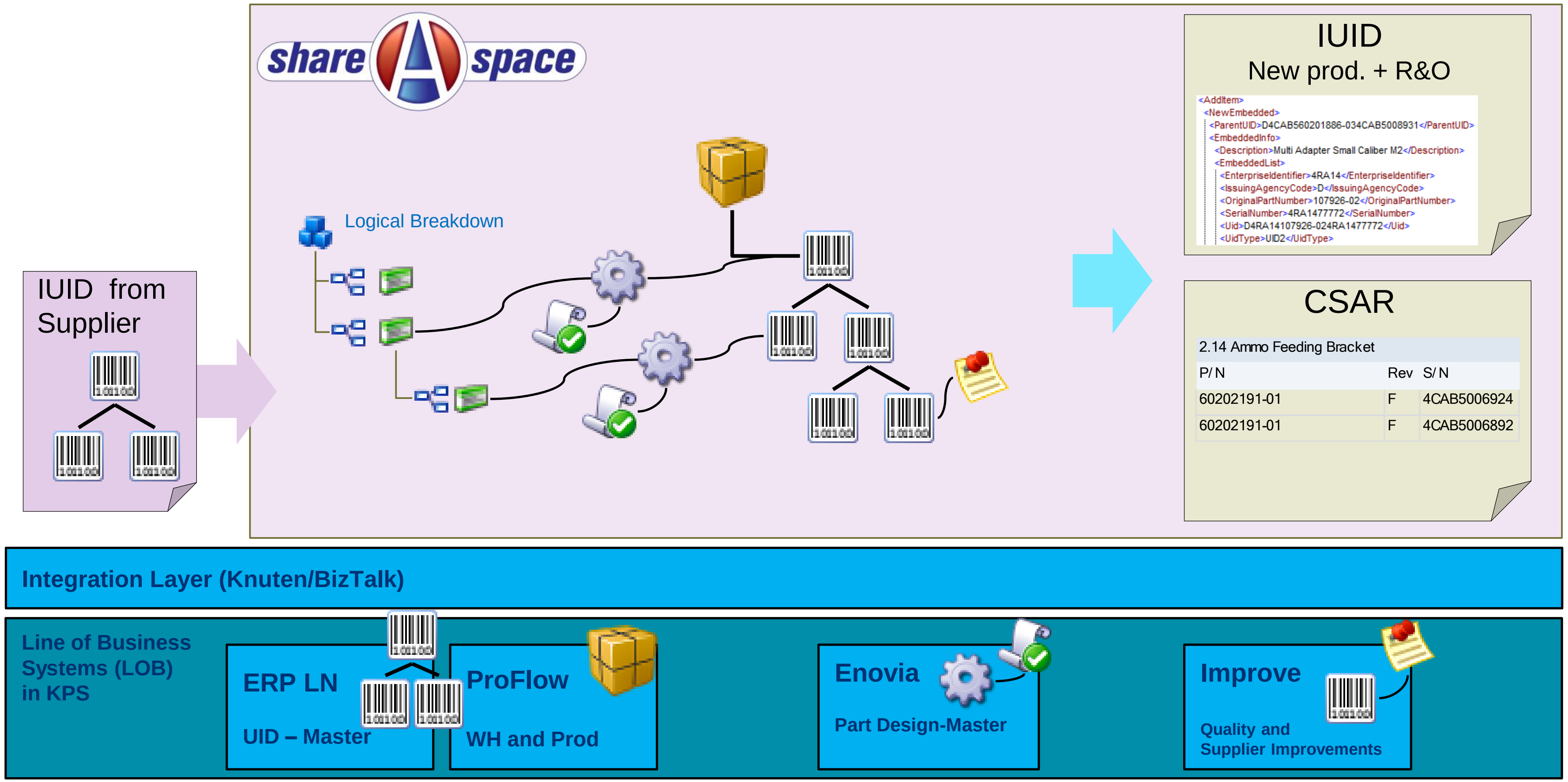
- Ensure good quality of delivered KPS CM and product data;
- Simplify the production of support data at KPS;
- Using a standard (ISO 10303-239, PLCS) for delivering KPS Norway support data to customers and others.

Includes the following tasks:

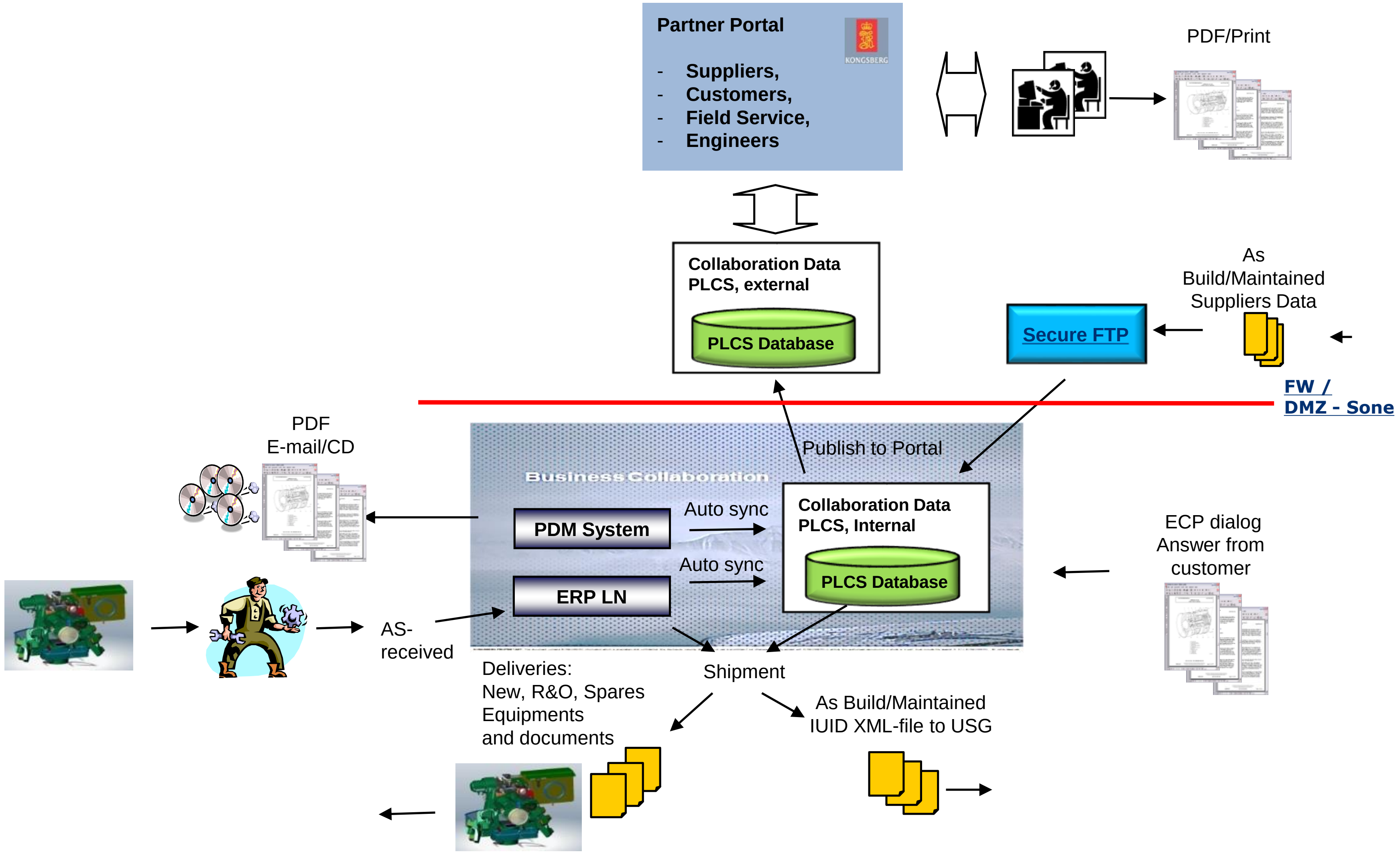
- Conversion and integration of product data from KPS systems (PDM/ERP) to PLCS;
- Automated report documenting the CM data
- Use of a PLCS end-user application

Product reporting tool

IUID & CSAR reporting solution for new production and R&O



Business Collaboration - Scope and information flow

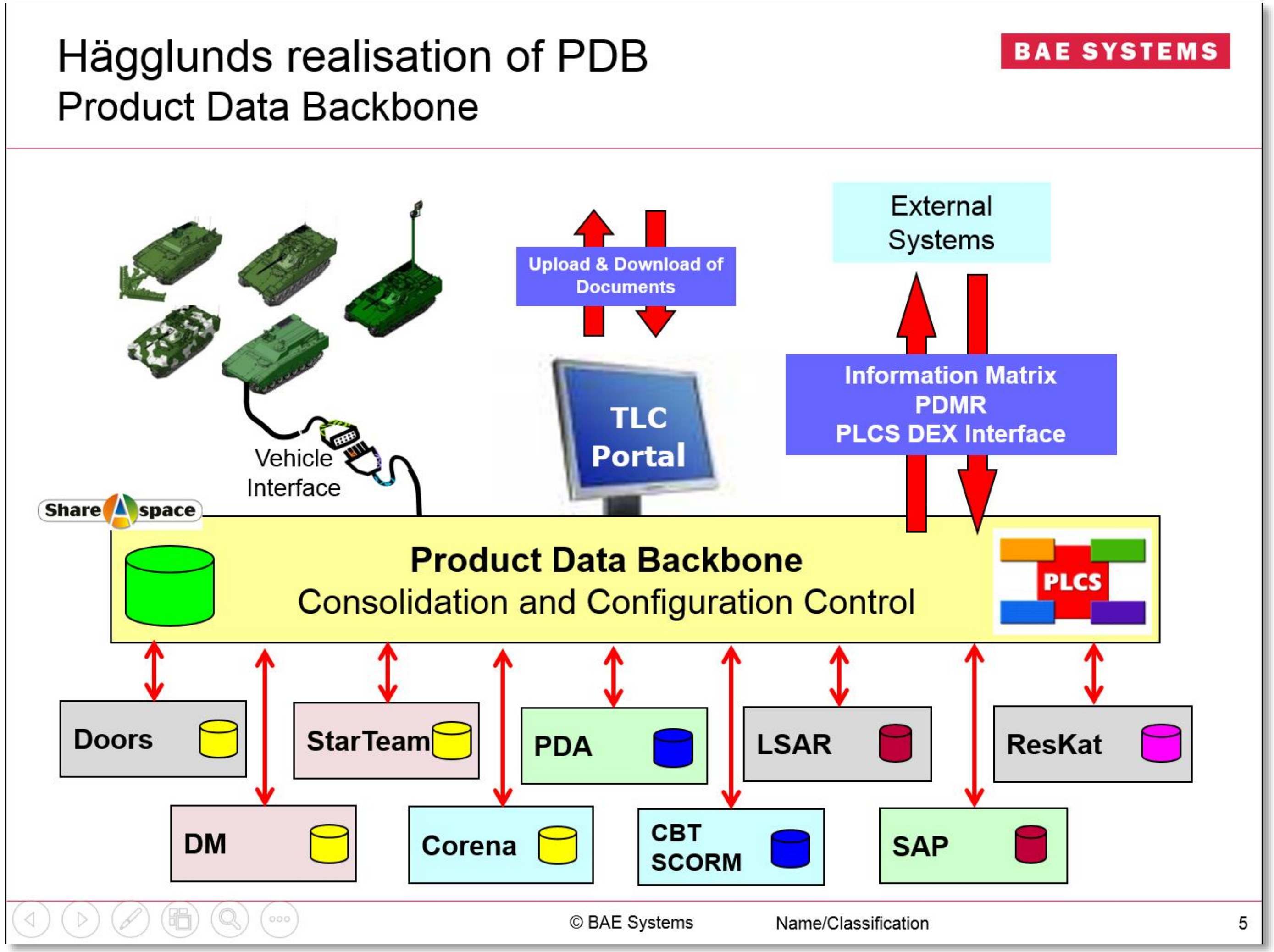


Kongsberg – Configuration Management

- Scope:
  - Configuration Management and External Collaboration through product lifecycle
- Challenge:
  - Share product data with customers and partners
  - Consolidation of as-design, as-built and as-maintained
- Solution:
  - Configuration Management repository and customer Collaboration Portal using Share-A-space®

- Business Benefit:
  - Improved quality in information deliveries
  - Reduced manual work creating information packages
  - More efficient partner and customer communication

BAE System Hägglunds Product Data Backbone



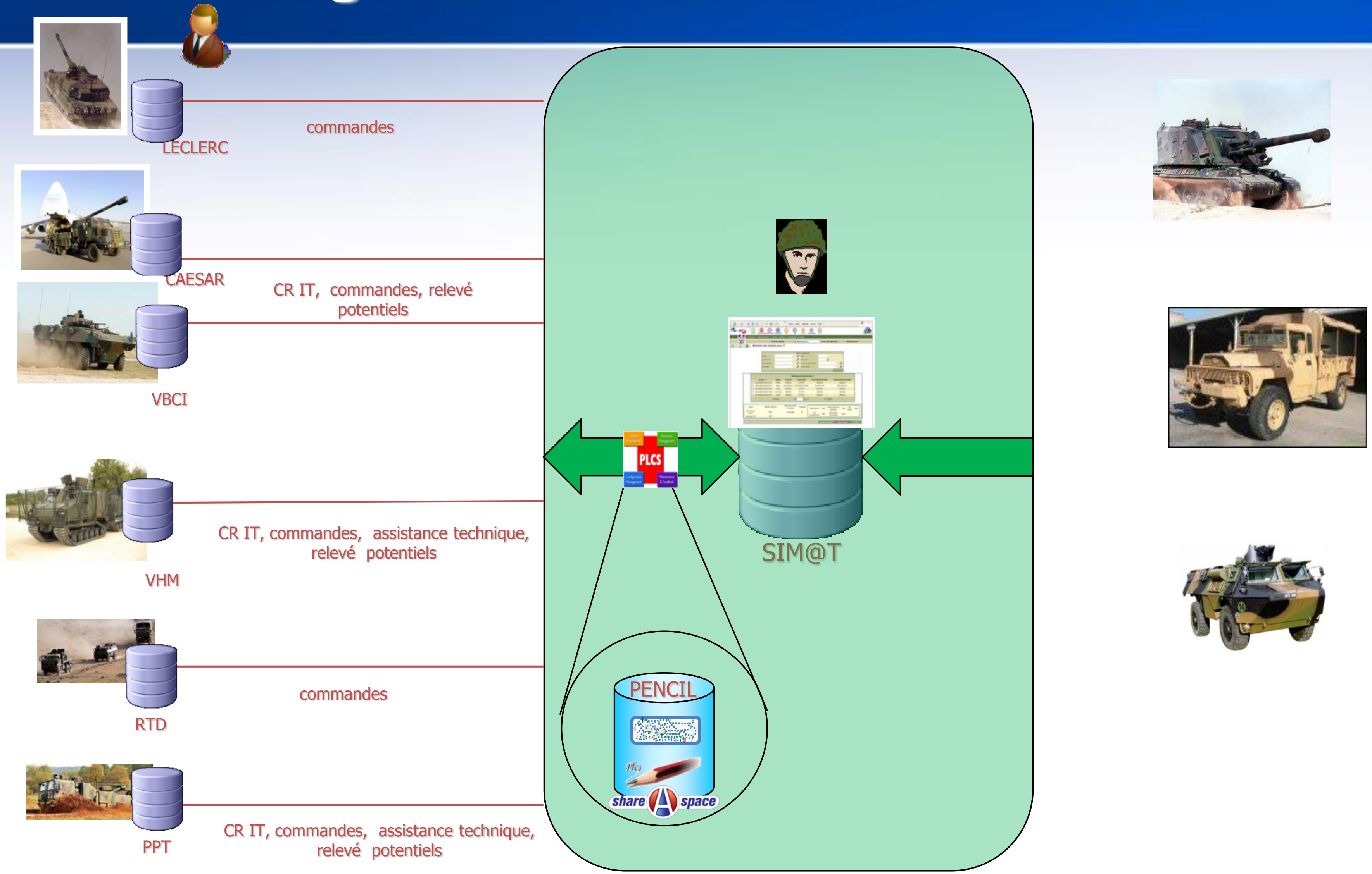
- Many interfaces of different kind
- Users use their traditional systems
- Information consolidated in the PDB
- Large data volumes

Scania – Higer PLM collaboration

- Scope:
  - Product data collaboration between OEM and Partner
- Challenge:
  - Share product data with partner
  - Outsourced body design and manufacturing
  - Consolidate as-built with as-designed and as-maintained
  - Feed back into Scania PLM and in-service platforms
- Solution:
  - Share-A-space portal to consolidate as-built with Scania model programs

- Business Benefits:
  - Reduced manual input data
  - Improved data quality over lifecycle
  - High traceability as-designed, as-built and as-maintained

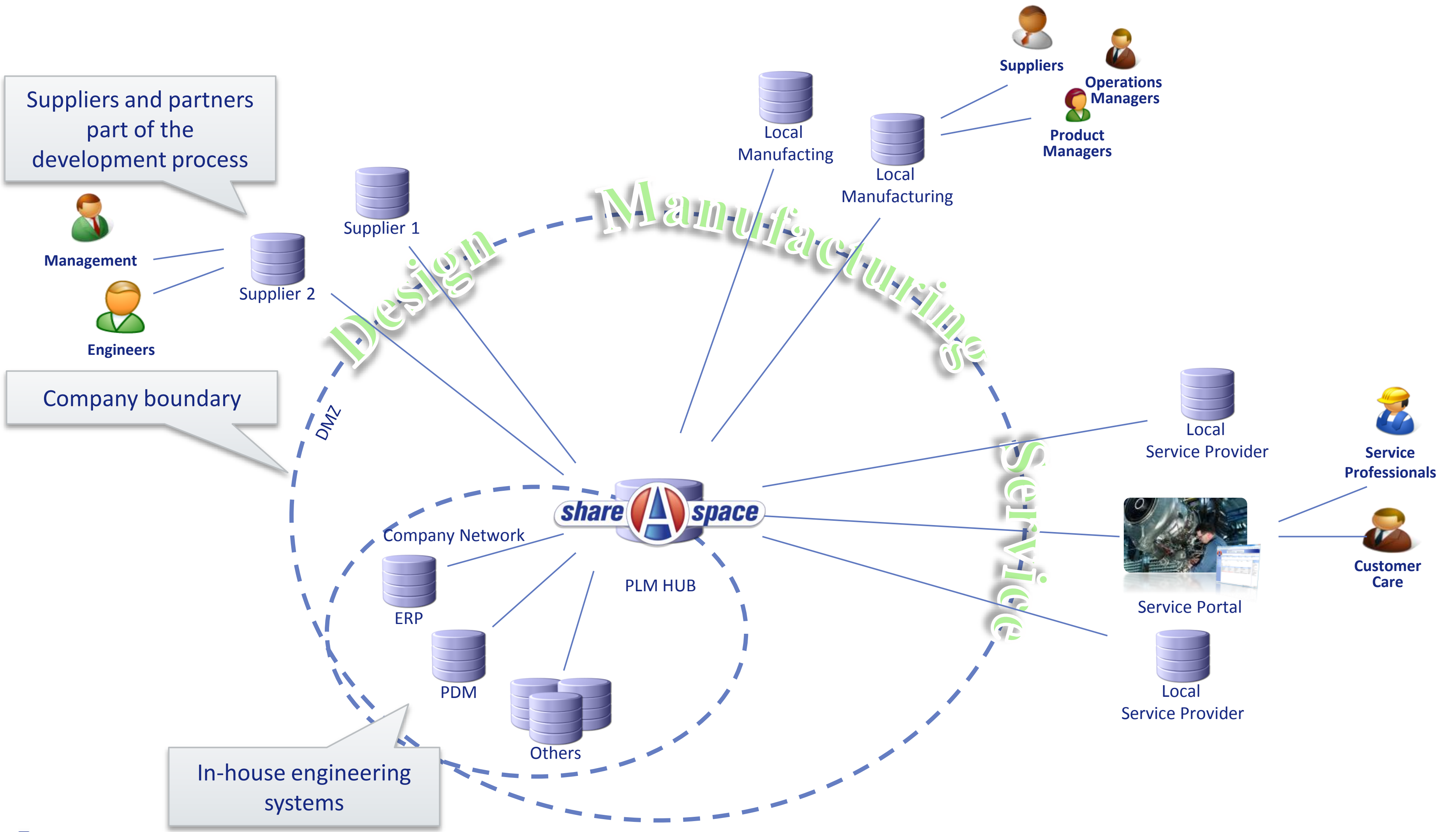
PENCIL – A hub
for sharing data between SIMMT and OEMs



Structure intégrée du maintien en condition opérationnelle des matériels terrestres

PENCIL : Plateforme d'Echange Normalisé et Centralisé d'Information Logisitique

What else is needed beyond a standard?



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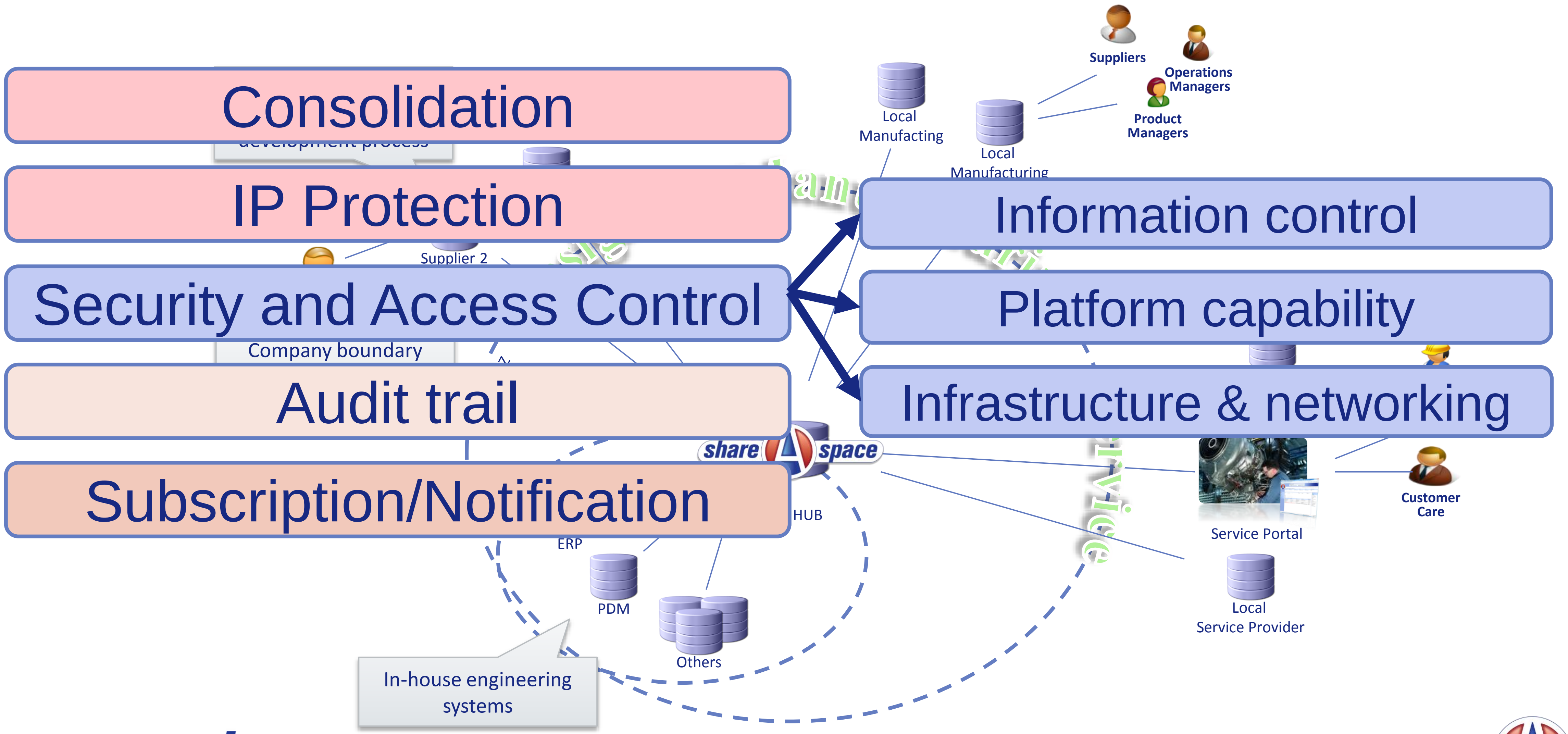

Integrating PLM, SE and CM for lean, innovative and agile operations

Rich model

Flexible

PLCS

Designed to fit with AP233 for SE

Proven to work in different industries

IT Independent

Process Neutral


T H A N K
Y O U

Sources

- Full standard
  - ISO.ch
- DEXs as OASIS standard
  - http://docs.oasis-open.org/plcs/dexlib/cs01/oasis_cover.htm
- DEXs development area
  - http://www.plcs.org/plcslib/plcslib/